# Sports Tournament Management System

**Data Engineering Capstone Project – 2026**
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## 1. Problem Statement

Managing sports tournaments manually is inefficient and error-prone. Organizers often face challenges such as scheduling conflicts, inconsistent record-keeping, difficulty in tracking player performance, and lack of analytical insights. With multiple tournaments running simultaneously across different sports, maintaining structured and accurate data becomes increasingly complex.

This project aims to solve these issues by designing a **centralized database system** that can efficiently manage tournaments, teams, players, matches, and venues. The system ensures data consistency, reduces redundancy, and enables advanced analytical queries for better decision-making.

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## 2. Solution Overview

The **Sports Tournament Management System** is a relational database solution built using MySQL. It provides a structured approach to manage all aspects of a sports tournament ecosystem.

The system is designed using **normalized database principles (up to 3NF)** to ensure minimal redundancy and maximum data integrity. It supports multiple sports such as Football, Cricket, Basketball, and others within a single unified database.

The solution includes:

- Structured database schema with relationships
- Automated updates using triggers
- Business logic encapsulated using stored procedures
- Analytical insights using advanced SQL queries

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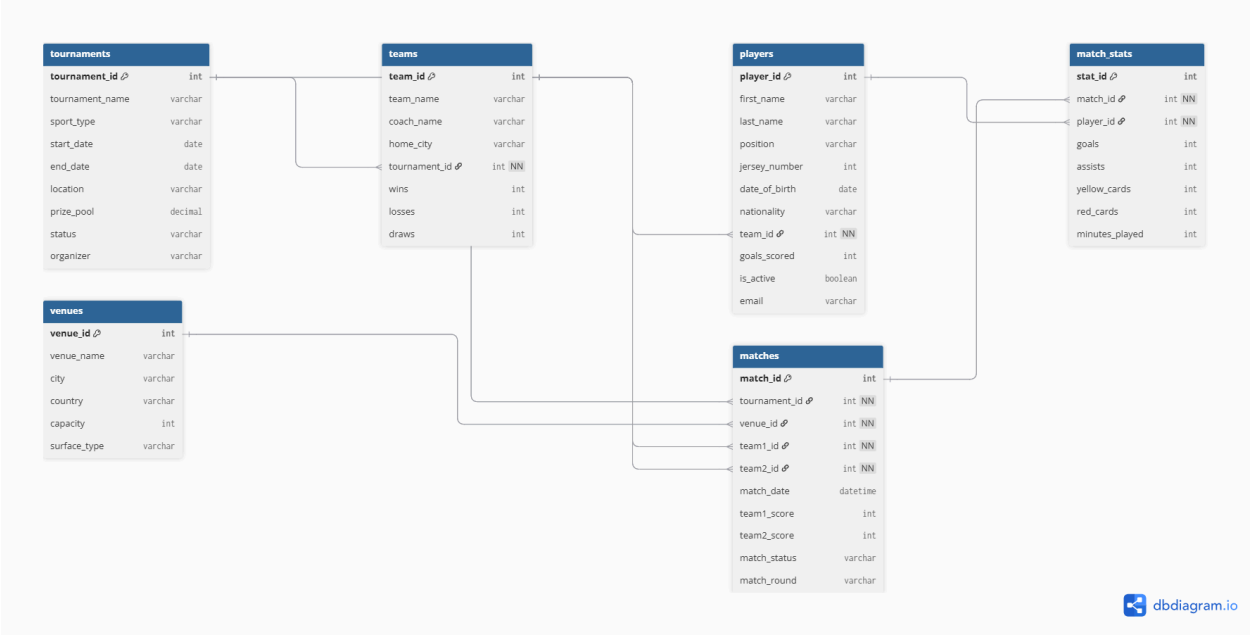
## 3. Database Design & ER Diagram

The system consists of six main entities:

- **Tournaments** – Stores tournament details such as name, sport type, and schedule
- **Teams** – Represents teams participating in tournaments
- **Players** – Stores player information linked to teams
- **Matches** – Contains match schedules, scores, and results
- **Venues** – Stores location and capacity details
- **Match_Stats** – Captures player performance per match

**Relationships:**

- One tournament has multiple teams and matches
- One team has multiple players
- One match involves two teams
- One match has multiple player statistics
- One player can have multiple match statistics

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## 4. Key Features

### Database Features

- Fully normalized schema (3NF)
- Primary and foreign key constraints
- Check constraints for data validation
- Support for multiple sports

### SQL Implementation

- DDL operations: CREATE, ALTER, DROP
- DML operations: INSERT, UPDATE, DELETE
- 20+ records per main table

### Advanced SQL Features

- **Views** for simplified data access
- **Indexes** for performance optimization
- **Stored Procedures** for business logic
- **Functions** for reusable calculations
- **Triggers** for automation

### Business Analytics

- Tournament standings
- Top player analysis
- Venue utilization insights
- Match trends and performance metrics

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## 5.Query-Outputs

**Output 1 — Tournament Standings (Premier Football League 2026)**

| Rank | Team Name | W | D | L | Played | Points |
|---|---|---|---|---|---|---|
| 1 | Mumbai Strikers | 9 | 2 | 2 | 13 | 29 |
| 2 | Delhi Dynamos | 7 | 2 | 3 | 12 | 23 |
| 3 | Bangalore Blasters | 6 | 2 | 4 | 12 | 20 |
| 4 | Kolkata Knights | 5 | 2 | 5 | 12 | 17 |
| 5 | Chennai Challengers | 4 | 2 | 6 | 12 | 14 |
| 6 | Hyderabad Hawks | 3 | 2 | 7 | 12 | 11 |

*ROW_NUMBER() OVER (PARTITION BY tournament_id ORDER BY points DESC) from vw_tournament_standings*

**Output 2 — Top 5 Goal Scorers**

| Player Name | Team | Tournament | Goals |
|---|---|---|---|
| Aryan Mehta | Mumbai Strikers | Premier Football League 2026 | 13 |
| Akash Kumar | Chennai Challengers | Premier Football League 2026 | 11 |
| Aditya Singh | Delhi Dynamos | Premier Football League 2026 | 10 |
| Manish Tiwari | Kolkata Knights | Premier Football League 2026 | 9 |
| Suresh Rao | Bangalore Blasters | Premier Football League 2026 | 8 |

*SELECT from vw_top_scorers ORDER BY goals_scored DESC LIMIT 5*

**Output 3 — Venue Utilization Report**

| Venue | City | Capacity | Total Matches | Completed | Upcoming |
|---|---|---|---|---|---|
| Eden Gardens | Kolkata | 68,000 | 8 | 6 | 2 |
| Wankhede Stadium | Mumbai | 33,000 | 6 | 5 | 1 |
| M. Chinnaswamy | Bangalore | 40,000 | 6 | 4 | 2 |
| Jawaharlal Nehru | Delhi | 75,000 | 5 | 3 | 2 |
| YMCA Ground | Chennai | 10,000 | 2 | 1 | 1 |

*venues LEFT JOIN matches GROUP BY venue_id — identifies overused/underused venues for logistics planning*

**Output 4 — Player Discipline Report**

| Player | Team | Yellow Cards | Red Cards | Discipline Status |
|---|---|---|---|---|
| Rohan Sharma | Mumbai Strikers | 2 | 0 | ■ Caution |
| Aditya Singh | Delhi Dynamos | 2 | 0 | ■ Caution |
| Aryan Mehta | Mumbai Strikers | 0 | 0 | ✓ Clean |
| Pradeep Nair | Delhi Dynamos | 1 | 0 | ✓ Clean |
| Manish Tiwari | Kolkata Knights | 1 | 0 | ✓ Clean |

*CASE: 3+ yellows=Suspension Risk | 2+=Caution | else=Clean (HAVING total_cards>0, ORDER BY yellows DESC)*

**Output 5 — Head-to-Head: Mumbai Strikers vs Delhi Dynamos**

| Round | Date | Score | Winner |
|---|---|---|---|
| Group Stage | Jan 15, 2026 | 3 – 1 | Mumbai Strikers |
| Semi Final | Feb 01, 2026 | 2 – 0 | Mumbai Strikers |
| Final | Mar 30, 2026 | 3 – 2 | Mumbai Strikers |

*CONCAT(team1_score,'–',team2_score) + CASE winner logic from vw_match_schedule view*

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## 6. Tech Stack

- **Database:** MySQL 8.0
- **Language:** SQL (DDL, DML, Advanced SQL)
- **Tools Used:** MySQL Workbench, dbdiagram.io, GitHub

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## 7. Unique Points

- Supports **multiple sports tournaments** in a single database
- Uses **advanced SQL features** beyond basic CRUD operations
- Implements **automation using triggers**
- Includes **real-world analytical queries** for decision-making
- Designed with **scalability and normalization in mind**

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## 8. Business Insights

The system provides valuable insights such as:

1. Tournament standings and rankings
2. Top goal scorers across tournaments
3. Head-to-head performance between teams

4. Venue utilization efficiency
5. Player discipline tracking (cards analysis)
6. Monthly match trends
7. Team performance and win percentage

These insights help organizers make better strategic and operational decisions.

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## 9. Future Improvements

- Develop a **web-based dashboard** for visualization
- Integrate **real-time match updates** using APIs
- Add **user authentication system** for admins and users
- Implement **data warehousing for historical analysis**
- Expand to support **international tournaments and leagues**

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## 10. Conclusion

The Sports Tournament Management System successfully demonstrates the application of database design principles and advanced SQL techniques in solving real-world problems. It provides a scalable, efficient, and structured solution for managing complex tournament data.

The project highlights strong understanding of:

- Database normalization
- Relational modeling
- SQL programming
- Analytical query design

This system can be further extended into a full-fledged application with frontend and backend integration.

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